

Education

Nanjing University Sep 2023 – present
B.Eng. in Intelligent Software Engineering (expected Jun 2027)
School of Intelligent Software and Engineering
Average: 88.4/100

Research Interests

AI systems that learn and improve continually from data, experience, and human feedback. Interests: **data-centric AI, self-improving agents, and human–AI interaction.**

Research Experience

The Hong Kong University of Science and Technology (Guangzhou)
Visiting Research Student Aug – Oct 2026
Research with Prof. Weikai Yang

Nanjing University, iMATE Lab Mar 2025 – present
Research Assistant | Advisor: Prof. Ke Xu
School of Intelligence Science and Technology; collaboration with Dr. Wenchao Li.

Publications and Preprints

NetworkCanvas: Supporting Progressive Network Visualization Exploration via Adaptive Recommendations

Wenchao Li, Yuewen Gao, **Yu He**, Cong Zhu, Ke Xu.
CHI 2026 | Published conference paper | DOI

SkillCommit: Evolving Agent Skills through Behaviorally Validated Scope Expansion

Yu He, Weikai Yang.
Submitted to AAAI 2027 | arXiv preprint, 2026 | Paper

MSSR: Memory-Aware Adaptive Replay for Continual LLM Fine-Tuning

Yiyang Lu, **Yu He**, Jianlong Chen, Hongyuan Zha.
arXiv preprint, 2026 | Paper

Manuscripts Under Review

SliceTune: Supporting Interactive Intent-Guided Slice Finding for Model Validation

Submitted to CHI 2027

SkillContour: Steering Agent Skill Consolidation with Behavioral Evidence

Submitted to CHI 2027

Selected Research Projects

SliceTune

- An interactive system translates validation goals into interpretable data attributes, guides failure-slice discovery, and connects diagnosis to repair planning.
- Evaluation includes a controlled study with 14 participants, three classification cases, and interviews with three experts.

SkillContour

- A mixed-initiative workbench helps people diagnose and revise agent skills using execution evidence, then replay and validate changes before commitment.
- Evaluation combines a 14-participant study with deployment of participant-reviewed skill libraries on 45 held-out NBA tasks.

SkillCommit

- A replay-validated algorithm turns agent experience into reusable skills, expanding their scope without updating model parameters.
- Improves agent performance on RuleArena, OpenExempt, and KOR-Bench; learned skills transfer across model sizes and families.

NetworkCanvas

Mar – Sep 2025

- Implemented visualization and feedback components for adaptive network exploration, including dataset integration and interaction design.
- Contributed to manuscript framing, figures, and experimental documentation.

MSSR

Aug 2025 – Jan 2026

- Modeled knowledge-retention dynamics and implemented adaptive replay scheduling to reduce forgetting in continual fine-tuning.
- Integrated datasets and pretrained models for evaluation across three models and 11 sequential tasks.

NewsTract

Jun 2025 – Jan 2026

- Designed data models and coordinated visualizations for exploring event connections across news broadcasts.
- Contributed to manuscript development and a 16-participant evaluation, including study design, analysis, and figures.

Selected Honors and Awards

Nanjing University People’s Scholarship	Nov 2025
Meritorious Winner, Mathematical Contest in Modeling Consortium for Mathematics and Its Applications (COMAP).	May 2025
National Third Prize, China Collegiate Innovation and Invention Competition	Dec 2024

Technical Skills

Programming: Python, TypeScript, JavaScript, Java, C++, SQL.
Machine learning: PyTorch; continual fine-tuning and agent evaluation.
Interactive systems: React, Vue 3, D3.js, G6.js, ECharts; Git/GitHub.
Languages: Chinese (native); English.